

Class Teacher _____ Name _____

SYDNEY TECHNICAL HIGH SCHOOL



Mathematics Extension 1

Year 11 Preliminary Course

Assessment Task 3

September 2014

Time Allowed: 90 minutes

General Instructions:

- Write using black or blue pen.
- Approved calculators may be used.
- Attempt all questions.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start each question on a new side of a page.

Total Marks 71

Section 1 – Multiple Choice 5 Marks Answer on sheet after question 5. Do not tear this sheet out. Allow 8 minutes for this section	Section 2 66 Marks Allow 82 minutes for this section
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SECTION 1 – MULTIPLE CHOICE (FILL IN YOUR ANSWERS ON THE ANSWER SHEET PROVIDED-DO NOT TEAR THE SHEET OUT)

1. A parabola has its focus at (0, 4). The equation of its directrix is $x = -4$.

Which of the following is the equation of the parabola?

- A. $x^2 = 16y$
- B. $(x + 2)^2 = 8(y - 4)$
- C. $(y + 2)^2 = 8(x - 4)$
- D. $(y - 4)^2 = 8(x + 2)$

2. Which one of the following expressions represents the factored form of $8x^3 + 27$?

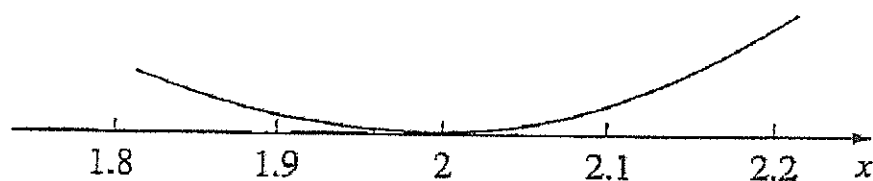
- A. $8x^3 + 27 = (2x + 3)(4x^2 + 6x + 9)$
- B. $8x^3 + 27 = (2x + 3)(4x^2 - 6x + 9)$
- C. $8x^3 + 27 = (2x - 3)(4x^2 - 6x - 9)$
- D. $8x^3 + 27 = (2x - 3)(4x^2 + 6x - 9)$

3. Consider the function $f(x) = \frac{x^4 + 3x^2}{x^4 + 3}$

Which one of the following statements is correct?

- A. $f(x)$ is odd and $\lim_{x \rightarrow \infty} f(x) = 1$
- B. $f(x)$ is even and $\lim_{x \rightarrow \infty} f(x) = 3$
- C. $f(x)$ is even and $\lim_{x \rightarrow \infty} f(x) = 1$
- D. $f(x)$ is odd and $\lim_{x \rightarrow \infty} f(x) = 3$

4. Part of the graph of $y = P(x)$, where $P(x)$ is a polynomial of degree four, is shown below.



Which of the following could be the polynomial $P(x)$?

- A. $P(x) = x^2(x + 2)^2$
B. $P(x) = (x + 2)^4$
C. $P(x) = x(x - 2)^3$
D. $P(x) = (x - 1)^2(x - 2)^2$

5. The normal to the graph of $y = \sqrt{b - x^2}$ has a gradient of 3 when $x = 1$.

The value of b is

- A. $-\frac{10}{9}$
B. $\frac{10}{9}$
C. 4
D. 10

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SECTION A: MULTIPLE CHOICE

Instructions:

- Circle the letter that best answers the question
- One mark each

- | | | | | |
|----|---|---|---|---|
| 1. | A | B | C | D |
| 2. | A | B | C | D |
| 3. | A | B | C | D |
| 4. | A | B | C | D |
| 5. | A | B | C | D |

SECTION 2

QUESTION 6 (start a new page)

Marks

- (a) Solve $\frac{2x+1}{x-1} > 3$ 2
- (b) (i) Sketch $y = x^2 - 1$ 1
(ii) Hence, on a separate diagram sketch $y = |x^2 - 1|$ 1
- (c) $P(x)$ is an odd monic polynomial of degree 3. 2
If $P(3)=0$, sketch the polynomial.
- (d) Differentiate $y = \frac{x+1}{\sqrt{x}}$ and express the derivative as a simplified fraction. 2
- (e) Use the substitution $t = \tan \frac{x}{2}$ to show that 3
$$\frac{1+\sin x}{1-\cos x} = \cot \frac{x}{2} + \frac{1}{2} \operatorname{cosec}^2 \frac{x}{2}$$

QUESTION 7 (Start a new page)

Marks

- (a) The equation $2x^2 + px + q = 0$ has one root three times the other.
Show that $3p^2 = 32q$. 2
- (b) For what values of k is $2x^2 - 5x + 4k$ positive definite? 2
- (c) A parabola has equation $y^2 + 8y = -12x + 8$
- (i) Find the coordinates of its vertex. 1)
- (ii) Sketch the parabola showing its x intercept. 1)
- (iii) On your sketch, display the focus and directrix. 2
- (d) Find the acute angle between the lines $x - \sqrt{3}y - 2 = 0$ and $\sqrt{3}x - y + 3 = 0$ 3

)

)

QUESTION 8 (Start a new page)

Marks

- (a) Show that the equation $x^2 + (k + 2)x + k = 0$ has two real roots for all real values of k . 2
- (b) Solve the equation $\cos 2x + 3\cos x + 2 = 0$ for $0^\circ \leq x \leq 360^\circ$ 3
- (c) (i) Show that $(x + 1)$ is a factor of $P(x) = x^3 - x^2 - 10x - 8$. 1
- (ii) Hence express $P(x) = x^3 - x^2 - 10x - 8$ as a product of three linear factors 1
- (iii) By sketching $P(x)$ or otherwise, solve the inequality $\frac{x^3 - 10x}{x^2 + 8} \geq 1$ 2
- (d) Find the domain and range of the function $f(x) = 3\sqrt{4 - x^2}$ 2

QUESTION 9 (Start a new page)

Marks

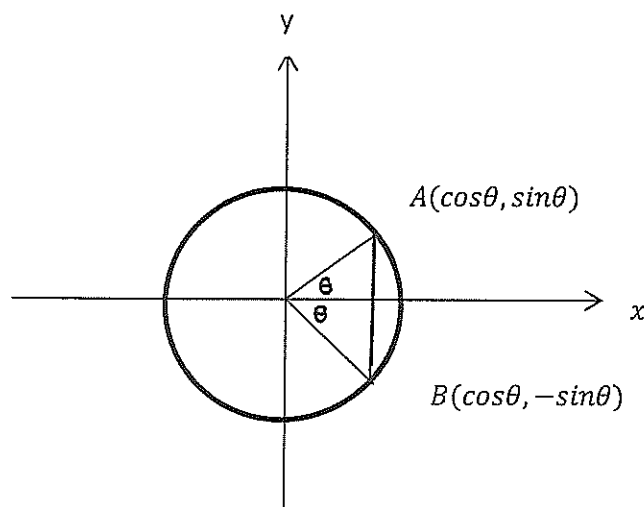
- (a) Simplify $\frac{1}{p^2-pq} - \frac{1}{pq-q^2}$ 2
- (b) The polynomial $P(x) = x^3 + a^2x^2 + ax + b$ leaves a remainder of 2 when divided by x and a remainder of 13 when divided by $x + 1$.
- (i) Show that $b = 2$ 1
- (ii) Find the value of a 2
- (c) (i) Express $\sin x + 3\cos x$ in the form $R \sin(x + \alpha)$ where $R > 0$ and $0^\circ < \alpha < 90^\circ$, giving the value of R in simplest exact form, and the value of α correct to the nearest degree. 2
- (ii) Solve the equation $3\cos x + \sin x + 2 = 0$ for $0^\circ \leq x \leq 360^\circ$, giving the solutions correct to the nearest degree. 2
- (d) Find the coordinates of point P on the curve $y = x\sqrt{x+3}$ where the tangent is parallel to the x - axis. 2

QUESTION 10 (Start a new page)

Marks

(a)

2



$A(\cos\theta, \sin\theta)$ and $B(\cos\theta, -\sin\theta)$, $0^\circ < \theta < 90^\circ$, are 2 points on the circle with centre at the origin and radius 1. Use the cosine rule in $\triangle AOB$ to show that $\cos 2\theta = 1 - 2\sin^2\theta$.

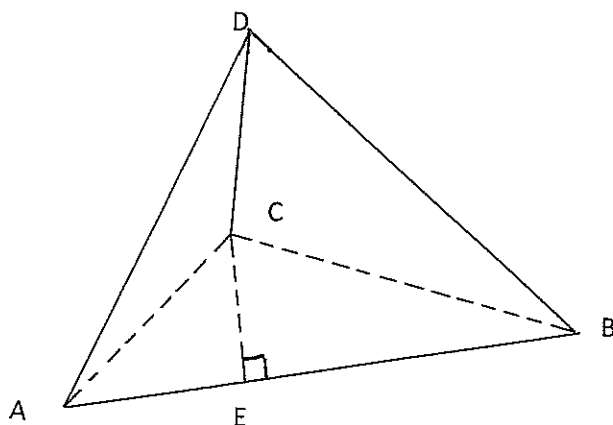
- (b) $A(8, \sqrt{50})$ and $B(1, \sqrt{18})$ are divided externally by a point P in the ratio of 3:1. Find the simplest exact form of this point.

3

- (c) Show that $\tan 75^\circ = 2 + \sqrt{3}$

2

(d)



CD is a vertical flagpole of height 10 metres. It stands with its base on horizontal ground. A and B are points on the ground due South and due East of C respectively. The angle of elevation of D is 45° from A and 30° from B. E is the foot of the perpendicular from C to AB.

- (i) Show that $\angle ABC = 30^\circ$
- (ii) Find the angle of elevation of D from E correct to the nearest minute.

2

2

QUESTION 11 (Start a new page)

Marks

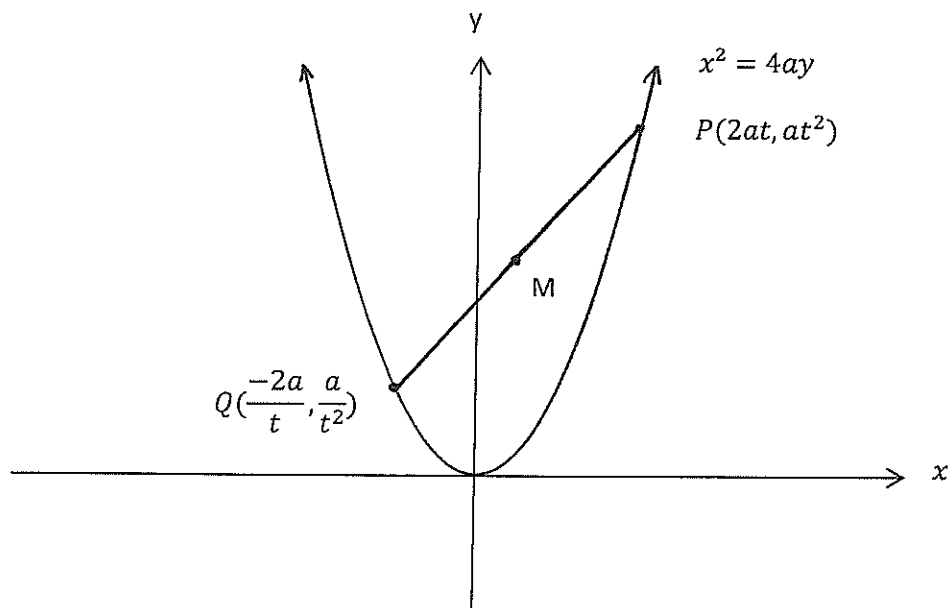
- (a) $P(x, y)$ is a variable point which moves in the number plane so that its distance from the point $A(3, 3)$ is twice its distance from the origin. Find the equation of the locus of P . 2

- (b) The polynomial $P(x) = x^3 + 2x^2 - 4x - 1$ has zeros α, β and γ so that
 $P(x) = (x - \alpha)(x - \beta)(x - \gamma)$.

- (i) Find the value of $(1 - \alpha)(1 - \beta)(1 - \gamma)$ 2
 (ii) Find the value of $(\beta + \gamma)(\gamma + \alpha)(\alpha + \beta)$ 2

- (c) Show that the equation of the normal at the point $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$ is $x + py = 2ap + ap^3$. 2

(d)



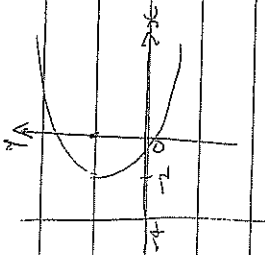
- $P(2at, at^2)$ and $Q(-\frac{2a}{t}, \frac{a}{t^2})$ are two points on the parabola $x^2 = 4ay$. M is the midpoint of the chord PQ . 3

As P and Q move on the parabola, find the locus of M .

END OF PAPER

2014 Ext 1 Final Prelim. Solutions

1. $(y \cdot)^2 = 8(x+2)$ 2. B



D

3. C 4. D 5. $\frac{dy}{dx} = \frac{1}{2}(b-x^2)^{-\frac{1}{2}} \cdot 2x$

$= -x$

$\sqrt{b-x^2}$

$\frac{-1}{2} = \frac{-1}{\sqrt{b-x^2}}$

$\frac{1}{2} = \frac{1}{b-x^2}$

D

6. a) $\frac{2x+1}{x-1} > 3$

$x \neq 1$

$\frac{2x+1}{x-1} = 3$

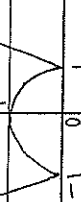
$2x+1 = 3x-3$

$4 = x$

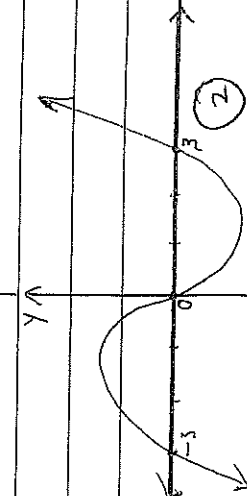


$1 < x < 4$ ②

①



c) $P(x) = x(x+3)(x-3)$



②

d) $y = \frac{x+1}{\sqrt{x}}$

$\frac{dy}{dx} = \frac{\sqrt{x} \cdot 1 - (x+1) \cdot \frac{1}{2\sqrt{x}}}{x}$

$= \frac{\sqrt{x} - \frac{x+1}{2\sqrt{x}}}{x} \cdot \frac{2\sqrt{x}}{2\sqrt{x}}$

$= \frac{2x - (x+1)}{2x^{\frac{3}{2}}}$

$= \frac{x-1}{2x^{\frac{3}{2}}}$ ②

$= \frac{x-1}{2x^{\frac{3}{2}}}$ ②

e) $\frac{1 + \sin x}{1 - \cos x} = \cot \frac{x}{2} + \frac{1}{2} \operatorname{cosec}^2 \frac{x}{2}$

LHS = $\frac{1 + \frac{2t}{1+t^2}}{1 - (\frac{1-t^2}{1+t^2})}$ ①

where $t = \tan \frac{x}{2}$

$= \frac{1+t^2+2t}{1+t^2-1+t^2}$

$= \frac{1+t^2+2t}{2t^2}$

$= \frac{1}{2t^2} + \frac{1}{2} + \frac{1}{t}$

$= \frac{1}{2} \left(\cot^2 \frac{x}{2} + 1 \right) + \frac{1}{\tan \frac{x}{2}}$ ①

$= \frac{1}{2} \operatorname{cosec}^2 \frac{x}{2} + \cot \frac{x}{2}$

$= \text{RHS}$ ①

$$7a) 2x^2 + 19x + 9 = 0$$

$$\text{Roots } d, 3d$$

$$4d = -\frac{19}{2} \quad 3d^2 = \frac{9}{2}$$

$$d = -\frac{19}{6}$$

$$d^2 = \frac{19^2}{36}$$

$$3 \times \frac{19}{64} = \frac{9}{2}$$

$$3p^2 = 329$$

$$b) 2x^2 - 5x + 4k$$

$$\Delta < 0$$

$$25 - 4 \times 2 \times 4k < 0$$

$$25 - 32k < 0$$

$$32k > 25$$

$$k > \frac{25}{32}$$

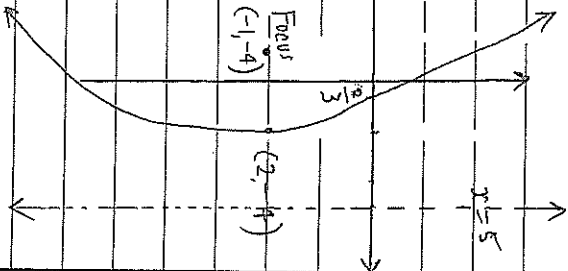
$$c) i) y^2 + 8y = -12x + 8$$

$$(ii)$$

$$y^2 + 8y + 16 = -12x + 24$$

$$(y+4)^2 = -12(x-2)$$

$$\text{Vertex } (2, -4)$$



$$d) x - \sqrt{3}y - 2 = 0$$

$$\sqrt{3}x - y + 3 = 0$$

$$y = \sqrt{3}x - \frac{2}{\sqrt{3}}$$

$$y = \sqrt{3}x + 3$$

$$m_1 = \sqrt{3}$$

$$m_2 = \sqrt{3}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\sqrt{3} - \sqrt{3}}{1 + \sqrt{3} \times \sqrt{3}} \right| = \frac{\sqrt{3} - \sqrt{3}}{2}$$

$$\theta = 30^\circ$$

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$$8g) x^2 + (k+2)x + k = 0$$

$$2) \text{ Real roots if}$$

$$\Delta > 0 \Rightarrow b^2 - 4ac > 0$$

$$(k+2)^2 - 4 \times 1 \times k > 0$$

$$k^2 + 4k + 4 - 4k > 0$$

$$k^2 + 4 > 0$$

$$\text{True for all } k$$

$$\therefore \text{ Two real roots.}$$

$$b) \cos 2x + 3 \cos x + 2 = 0$$

$$2 \cos^2 x - 1 + 3 \cos x + 2 = 0$$

$$2 \cos^2 x + 3 \cos x + 1 = 0$$

$$(2 \cos x + 1)(\cos x + 1) = 0$$

$$\cos x = -\frac{1}{2}, -1$$

$$x = 120^\circ, 240^\circ, 180^\circ$$

$$c) i) P(-1) = 0 \text{ need to show}$$

$$P(-1) = (-1)^3 - (-1)^2 + 10 - 8$$

$$= -1 - 1 + 10 - 8$$

$$= 0$$

$$\therefore (x+1) \text{ is a factor.}$$

$$= (x+1)(x+2)(x-4)$$

$$c) ii) \frac{x^3 - 10x}{x^2 + 8} \geq 1$$

$$x^3 - 10x \geq x^2 + 8$$

$$x^3 - x^2 - 10x - 8 \geq 0$$

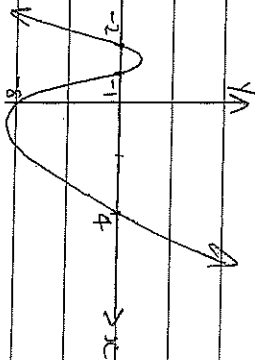
$$d) f(x) = 3\sqrt{4-x^2}$$

$$D: 4 - x^2 \geq 0$$

$$(2-x)(2+x) \geq 0$$

$$-2 \leq x \leq 2$$

$$R: 0 \leq y \leq 6$$



$$-2 \leq x \leq 2, x \geq 4$$

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9a. $\frac{1}{p(p-q)} - \frac{1}{q(p-q)} = 2 = b$
 $\therefore b = 2$

$= \frac{p^2 - q^2 - p^2 + pq}{pq(p-q)^2}$
 $= \frac{-(p^2 - 2pq + q^2)}{pq(p-q)^2}$
 $= \frac{-(p-q)^2}{pq(p-q)^2}$
 $= \frac{-1}{pq}$

cii) $R = \sqrt{12+3^2} = \sqrt{10}$
 $\tan \alpha = 3$
 $\alpha = 71^\circ 34'$
 $\sqrt{10} \sin(x+72) = -2$
 $\sin(x+72) = \frac{-2}{\sqrt{10}}$
 $x+72 = 2(9^\circ 14' 320' 46)$
 $x = 248^\circ 46', 147^\circ 14'$
 $x = 249^\circ, 147^\circ$

d) $y = x\sqrt{x+3}$
 $y' = x(x+3)^{\frac{1}{2}}$
 $y' = (x+3)^{\frac{1}{2}} + x \cdot \frac{1}{2}(x+3)^{-\frac{1}{2}}$
 $y' = \sqrt{x+3} + \frac{x}{2\sqrt{x+3}}$
 $y' = 0 = \frac{\sqrt{x+3}}{2\sqrt{x+3}} + \frac{x}{2\sqrt{x+3}}$ (Tangent parallel to x axis)
 $-\sqrt{x+3} = \frac{x}{2\sqrt{x+3}}$
 $x+3 = \frac{x}{4(x+3)}$
 $4(x+3)^2 = x^2$
 $4x^2 + 24x + 36 = x^2$
 $3x^2 + 24x + 36 = 0$
 $x^2 + 8x + 12 = 0$
 $(x+2)(x+6) = 0$
 $x = -2, -6$ not in domain

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 $(-2, -2)$

10a) $\cos 2\theta = \frac{1^2 + 1^2 - 4 \sin^2 \theta}{2}$ in $\triangle AOB$
 $\cos 2\theta = 1 - 2 \sin^2 \theta$

b) $x = \frac{-3 \pm \sqrt{1 + 1 \times 8}}{-3 \pm 1}, y = \frac{-3 \pm \sqrt{1 + 1 \times 8}}{-3 \pm 1}$
 $x = \frac{5}{-2}, y = \frac{9\sqrt{2} + 5\sqrt{2}}{-2}$
 $= \frac{-4\sqrt{2}}{-2}$
 $= 2\sqrt{2}$
 $(-2\frac{1}{2}, 2\sqrt{2})$

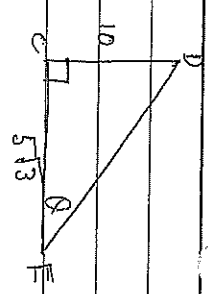
c) $\tan 75 = \tan(45+30)$
 $= \frac{\tan 45 + \tan 30}{1 - \tan 45 \tan 30}$
 $= \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} \times \frac{\sqrt{3}}{\sqrt{3}}$
 $= \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$
 $= \frac{4 + 2\sqrt{3}}{2}$
 $= 2 + \sqrt{3}$

d) Top View. AC is 10 ($\triangle ACD$ isosceles)
 $\tan 30 = \frac{10}{BC}$
 $\frac{1}{\sqrt{3}} = \frac{10}{BC} \therefore BC = 10\sqrt{3}$
 $\therefore \tan \angle ABC = \frac{10}{10\sqrt{3}}$
 $\angle ABC = 30^\circ$

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$$\text{civ) } \sin 30 = \frac{CE}{10}$$

$$\frac{1}{2} = \frac{CE}{10\sqrt{3}}$$



$$CE = 5\sqrt{3}$$

$$\tan \theta = \frac{10}{5\sqrt{3}}$$

$$= \frac{2}{\sqrt{3}}$$

$$\therefore \theta = 49.61^\circ$$

$$11. a) \sqrt{(x-3)^2 + (y-3)^2} = 2\sqrt{x^2 + y^2}$$

$$x^2 - 6x + 9 + y^2 - 6y + 9 = 4x^2 + 4y^2$$

$$0 = 3x^2 + 6x + 3y^2 + 6y - 18$$

$$0 = x^2 + 2x + y^2 + 2y - 6$$

$$b) \text{iv) } (1-\alpha)(1-\beta)(1-\gamma) = P(CD)$$

$$\therefore P(CD) = 1 + 2 - 4 - 1$$

$$= -2$$

$$\therefore (B+\gamma)(\gamma+\alpha)(\alpha+\beta)$$

$$= (-2-\alpha)(-2-\beta)(-2-\gamma)$$

$$= P(-2)$$

$$= (-2)^3 + 2(-2)^2 - 4(-2) - 1$$

$$= -8 + 8 + 8 - 1$$

$$= 7$$

$$c) y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a}$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

At $x=2ap$, m of tangent

is p $\therefore$ gradient of

Normal is $-\frac{1}{p}$

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$x + py = 2ap + ap^3$$

$$d) M \text{ is } \left(\frac{2a + \frac{a}{p}}{2}, \frac{a + 2 + \frac{a}{p^2}}{2} \right)$$

$$x = a + \frac{a}{p}$$

$$y = a + \frac{a}{p^2}$$

$$x^2 = a^2 \left(1 + \frac{1}{p^2} + \frac{1}{p^2} \right)$$

$$x^2 = a^2 \left(\frac{2p^2 + 1}{p^2} \right)$$

$$x^2 = 2ay - 2a^2$$

as reqd. from y equation

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